

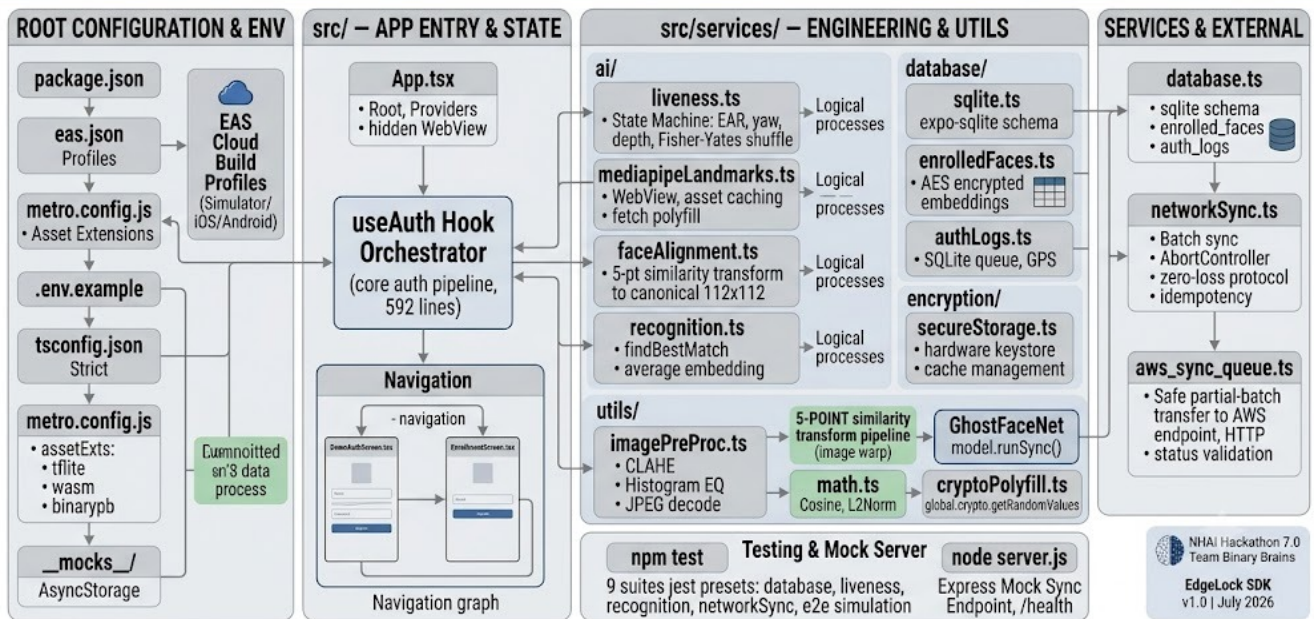
EDGE LOCK SDK

Visual Technical Deck

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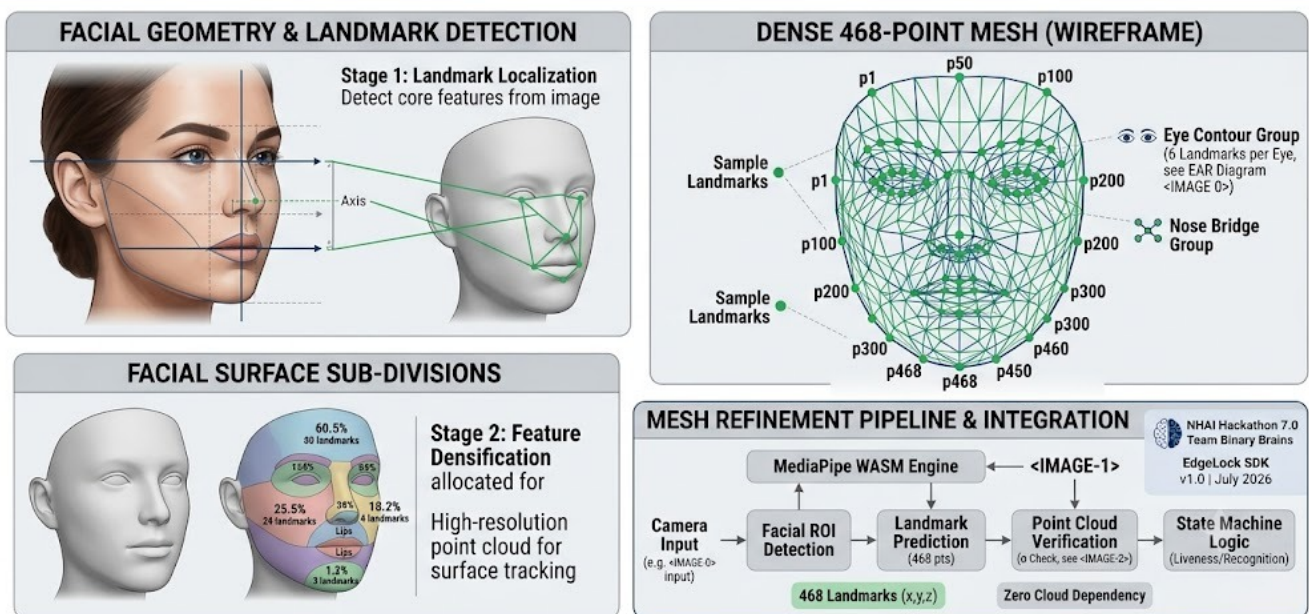
Offline Facial Recognition with Active Liveness & Zero-Loss Sync

EDGE LOCK SDK: FULL REPOSITORY CODEBASE & ARCHITECTURE MAP (NHAI Hackathon 7.0 Team Binary Brains | July 2026)



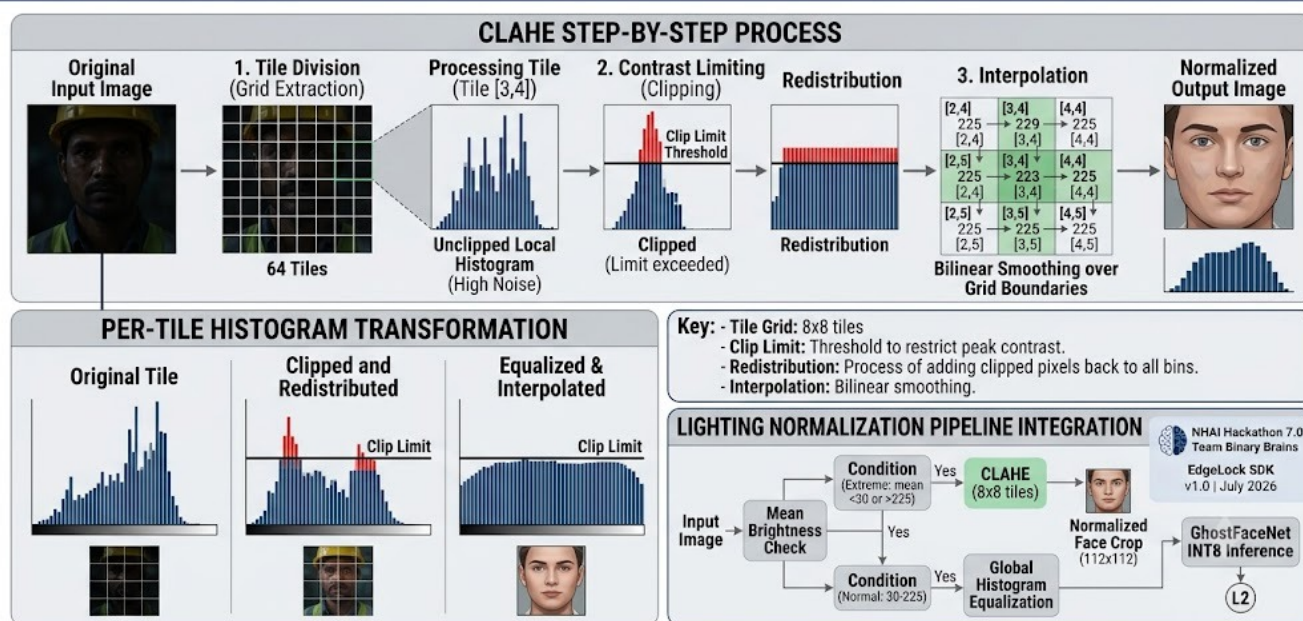
Full Repository Architecture & Codebase Map

EDGE LOCK SDK: MEDIAPIPE FACE MESH LANDMARK BREAKDOWN (468-POINT DENSE MESH)



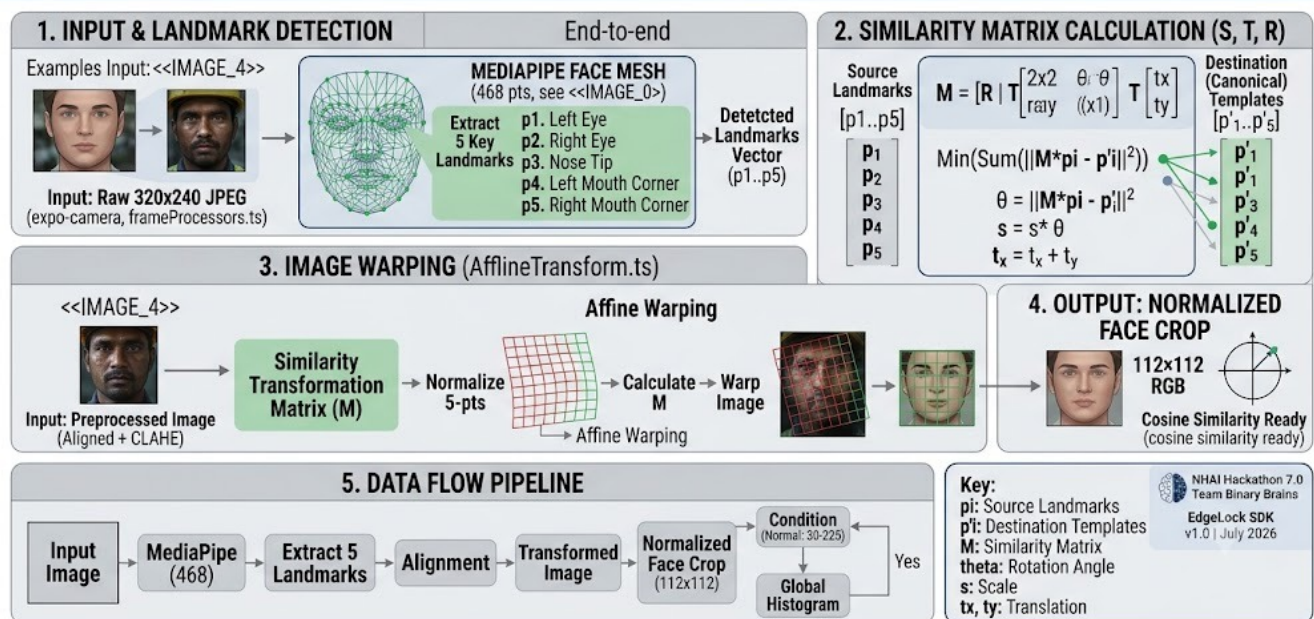
MediaPipe Face Mesh — 468-Point Dense Landmark Detection

EDGE LOCK SDK: CLAHE LIGHTING NORMALIZATION (imagePreProc.ts)



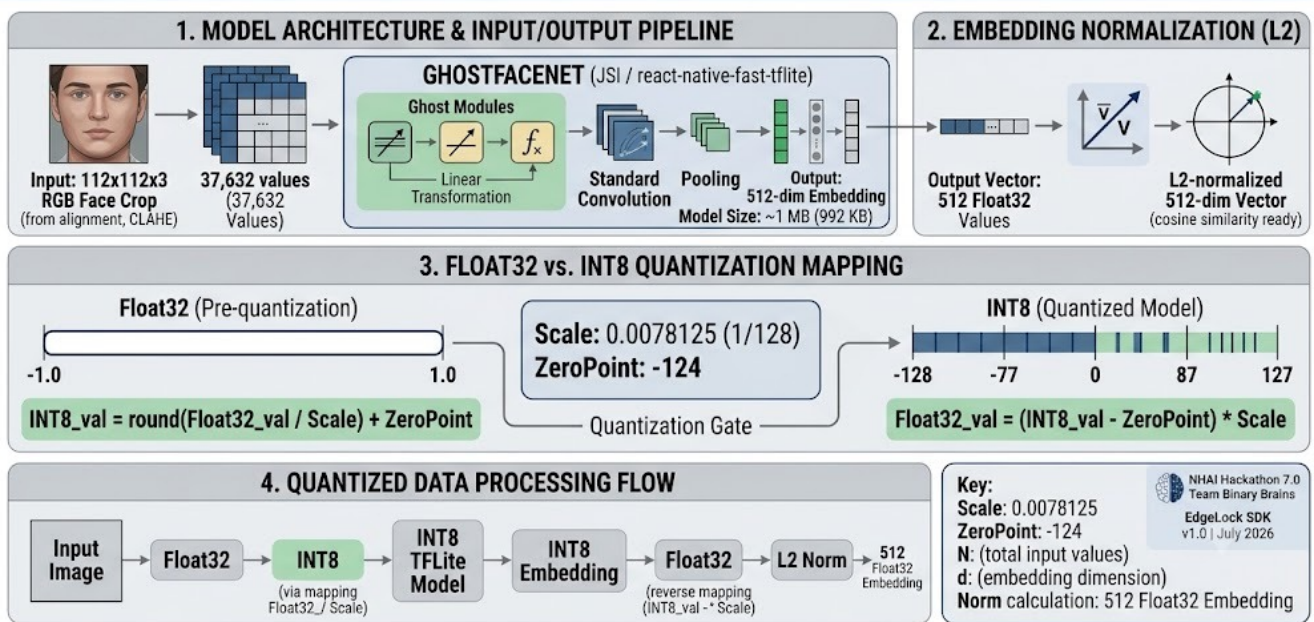
CLAHE Lighting Normalization — Harsh Field Conditions

EDGE LOCK SDK: 5-POINT FACE ALIGNMENT (faceAlignment.ts)



5-Point Face Alignment — Affine Warp to Canonical Crop

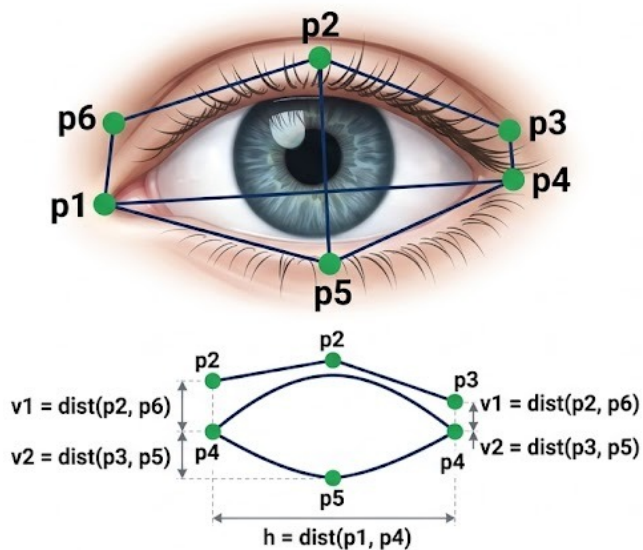
EDGE LOCK SDK: GHOSTFACENET ARCHITECTURE & INT8 QUANTIZATION (recognition.ts)



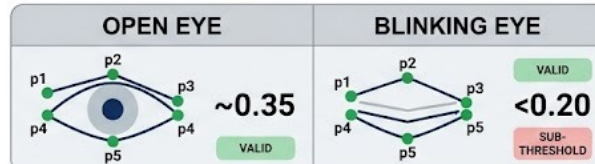
GhostFaceNet INT8 Quantized — 512-dim Embedding Extraction

EDGE LOCK SDK: EYE ASPECT RATIO (EAR) CALCULATION & LANDMARKS

6 key MediaPipe landmarks



$$\text{EAR} = \frac{\|p2 - p6\| + \|p3 - p5\|}{2 * \|p1 - p4\|}$$



Key:

p1 Inner Corner
 p2, p3 Upper Eyelid
 p4 Outer Corner
 p5, p6 Lower Eyelid

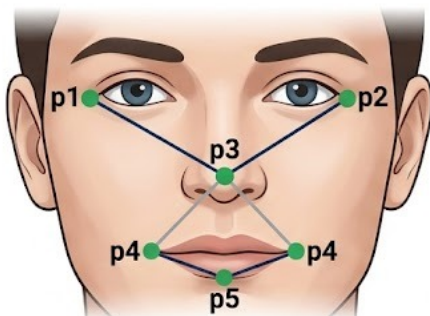
State Machine Logic



Eye Aspect Ratio (EAR) — Blink Detection State Machine

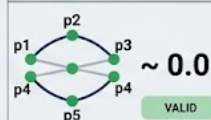
EDGE LOCK SDK: HEAD YAW CALCULATION & LANDMARKS

5 key MediaPipe landmarks

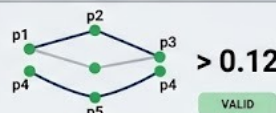


$$YAW = \frac{dL - dR}{dL + dR}$$

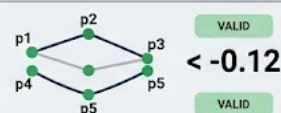
LOOKING STRAIGHT



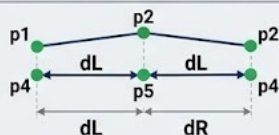
TURNING RIGHT



TURNING LEFT

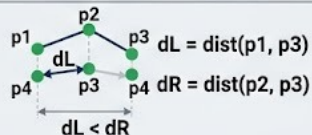


LOOKING STRAIGHT



LOOKING STRAIGHT

TURNING RIGHT

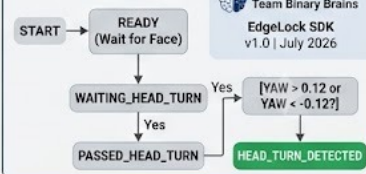


TURNING RIGHT

Key:

p1: Outer Eye Corner (Left)
 p2: Outer Eye Corner (Right)
 p3: Nose Tip (Center)
 p4: Mouth Corner (Left)
 p5: Mouth Corner (Right)
 dL, dR: Distance (Left/Right)
 Positive Yaw: Right Turn
 Negative Yaw: Left Turn

State Machine Logic

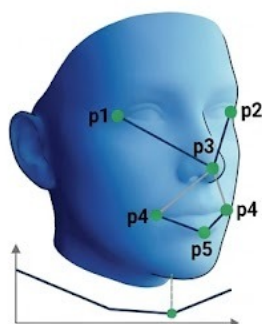


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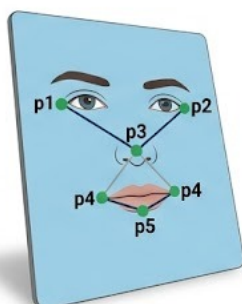
Head Yaw Calculation — Head Turn Detection State Machine

EDGE LOCK SDK: Z-COORDINATE STANDARD DEVIATION (Passive 3D Liveness Detection)

REAL 3D FACE



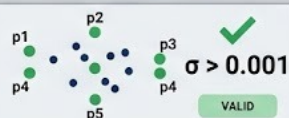
2D PRINT / SCREEN



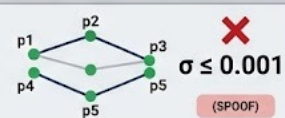
$$\sigma = \sqrt{\frac{1}{N} \sum_i^n (z_i - \mu)^2}$$

N : z-coouation of landmark i
 z_i : Total landmark
 μ : Standard coordinate

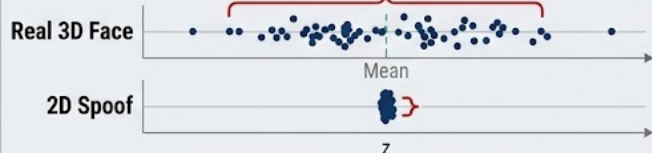
REAL FACE



SPOOF ATTACK



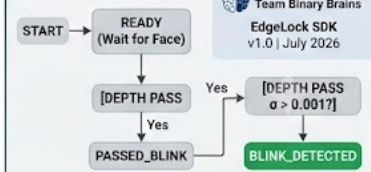
Depth Landmark Scatter (z-coord) with Standard Deviation



Key:

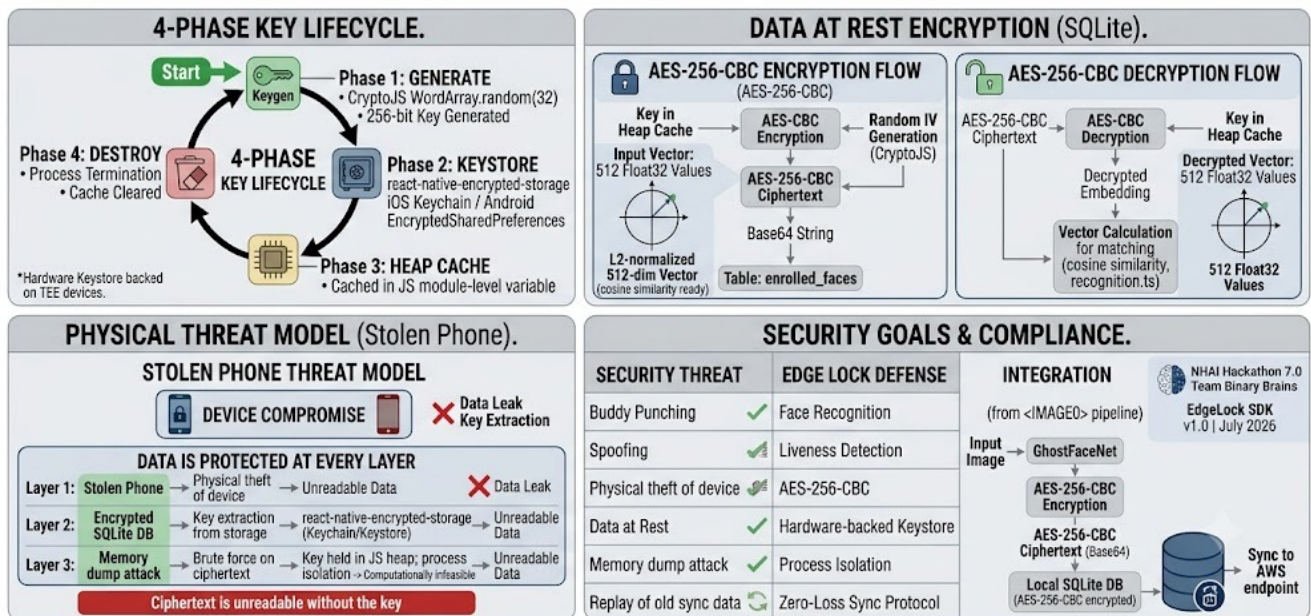
z_i : z-coordinate of landmark i
 N : Total Landmarks (468)
 μ : Mean z-coordinate
 σ : Standard Deviation (σ_{depth})
 Threshold: 0.001

State Machine Logic

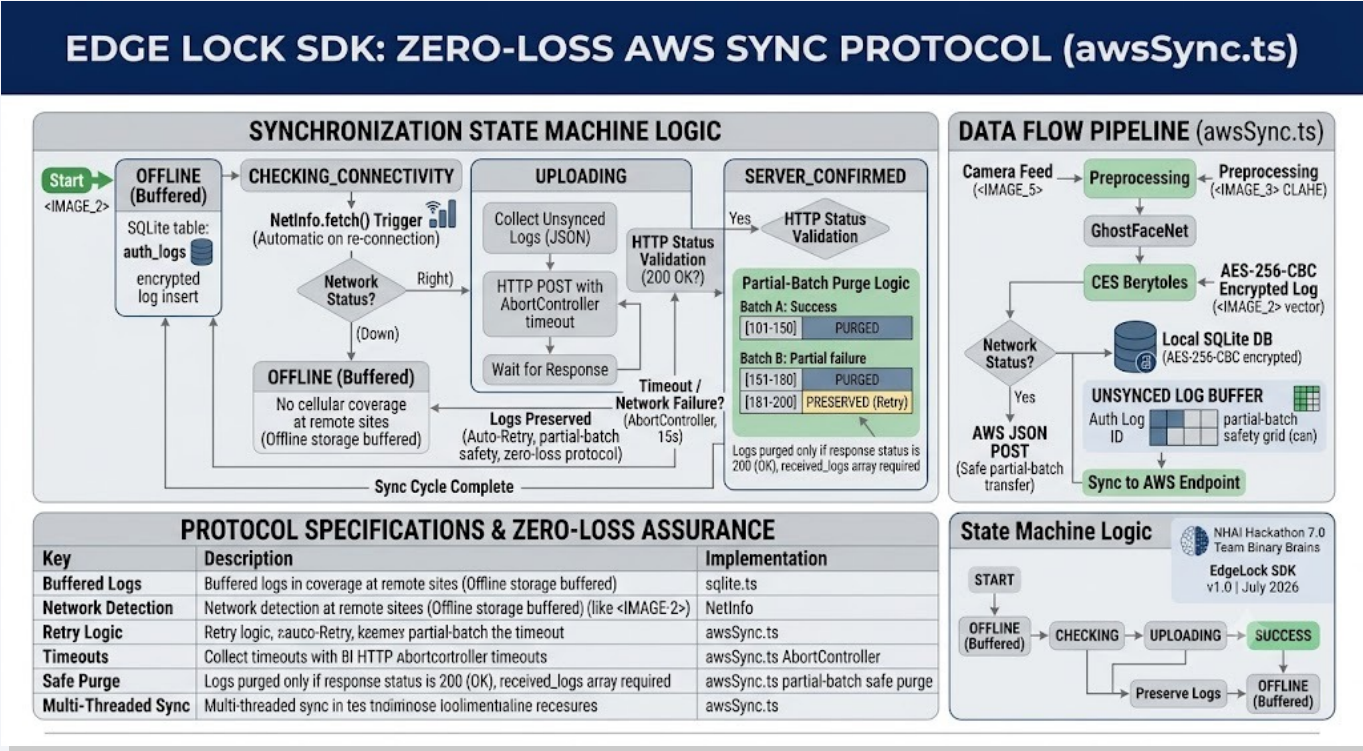


Z-Coordinate Standard Deviation — Passive 3D Spoof Detection

EDGE LOCK SDK: SECURITY ARCHITECTURE & KEY LIFECYCLE (AES-256-CBC, SQLite)



AES-256-CBC Security Architecture & 4-Phase Key Lifecycle



Zero-Loss AWS Sync Protocol — Offline-to-Cloud Pipeline

THANK YOU

Team Binary Brains | NHAH Hackathon 7.0

React Native + Expo SDK 50 | GhostFaceNet INT8 | AES-256-CBC | Zero-Loss Sync

Questions?